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1.0 INTRODUCTION

As the food and beverage sector becomes increasingly competitive, businesses need to leverage data-driven decision-making to optimize operations, tailor customer offerings, and drive business success (Dataintel, 2025). **Business Intelligence (BI)** has emerged as a key tool for the organizations that can convert raw data into meaningful information for strategic and operational decisions. By leveraging BI tools like Microsoft Power BI, businesses can gain insights from past sales data, track important metrics (KPIs), identify emerging trends and make data-driven decisions.

This report centers on one pizza restaurant called The Pizza Factory, which wants to enhance its business performance by introducing Business Intelligence solutions. The restaurant industry is rapidly changing and for restaurant operators to keep up with customer demand and expectations it is more important than ever to have knowledge of purchasing behavior, product performance, sales trends and operational efficiency. Data analytics and interactive dashboards provide management with valuable insights that can help them plan for marketing strategies, optimize inventory levels, enhance menu offerings, and boost profitability (Pathan et al., 2023; Tenzo, n.d.).

Data from the Pizza Sales dataset is used in the project, which provides detailed information about each transaction made, such as order details, product details, sales revenue, pizza category, pizza sizes, order time, etc. Data has been cleansed and transformed in Microsoft Power BI through the Power Query to ensure consistency and quality of the data before analysis. Different visualization techniques and Business Intelligence dashboards have been created to deliver actionable insights into the company's sales performance and customers' buying behavior.

The main goal of this report is to illustrate how Business Intelligence can help in the decision-making process at The Pizza Factory, by transforming the basic transactional data into interactive visualization reports. The report will include the background of the company, the data set used for analysis, data preparation, dashboard visualizations and analytics results, and the recommendations that can be used to help management improve operational and strategic performance.

2.0 COMPANY BACKGROUND

The Pizza Factory is a contemporary pizza restaurant, which offers a large selection of fresh, high-quality pizzas to its customers. The restaurant has a wide selection of pizza varieties, such as Classic, Supreme, Veggie, and Chicken pizzas, which come in various sizes to suit customer preferences. Along with product quality, The Pizza Factory strives to provide great customer service, run a smooth operation, and constantly improve the company to stay competitive in the food service industry.

The Pizza Factory is no different from many restaurants in that the business makes lots of sales transactions daily. Customer orders, product selections, quantities ordered, order dates and times, pizza sizes, pizza prices, total sales revenue, and quantities ordered. This data is gathered via the restaurant's point-of-sale (POS) system, but it cannot give any meaningful business insights without analysis. With the company continuing its growth, the management now need a better system to track business performance and opportunities for improvement.

Implementing Business Intelligence also helps operational managers to recognize patterns that might not be evident from the traditional reports. For instance, the management may know which types of pizzas make the maximum amount of money, which products aren't selling well, how many pizzas are sold on a particular day or week, and what can be anticipated on a particular day or week. These insights help the company optimize inventory management, manage employee schedules, tailor marketing strategies, and make informed price decisions.

Moreover, Business Intelligence enables strategic planning with reliable and timely information that enables to optimize the performance of an organization. Managers can make informed decisions, based on facts, that will minimize risk and maximize profits in their operations.

Overall, The Pizza Factory recognizes the importance of leveraging Business Intelligence as part of its digital transformation strategy. Through analyzing past sales information and providing visually engaging and informative visualizations via dashboards, the company can gain a better understanding of its business, improve customer satisfaction, optimize its operations, and sustain growth in a fiercely competitive restaurant landscape.

3.0 DATASET DESCRIPTION

The data for this project is a data set that contains the details of pizza sales transactions from the daily business of The Pizza Factory. The data has **48,620 rows and 12 columns**, where each row shows information about a single pizza that was ordered by a customer. Information contains **Pizza ID, Order ID, Pizza Name ID, Quantity, Order Date, Order Time, Unit Price, Total Price, Pizza Size, Pizza Category, Pizza Ingredients, Pizza Name**.

These properties are used to provide complete information that enables Business Intelligence analysis. The date of the transaction, the time of the order, the quantity, the unit price and the total price of the transaction information can help identify sales trends, peak business hours and analyze the performance of sales and revenue. The management can use product attributes, such as pizza category, pizza size, pizza name, and ingredients to assess the popularity of the product and customer preferences for various menu items.

The data was preprocessed, cleaned and transformed in Power Query of Microsoft Power BI for consistency and accuracy prior to analysis. The cleaned dataset was subsequently used to develop interactive dashboards that provide valuable insights into sales performance, product demand, and operational trends. These insights can help management make informed business decisions that will boost efficiency, improve profit and increase customer satisfaction.

3.1 Data Source

The pizza sales dataset used for this project is a publicly available dataset on **Kaggle**, which makes it suitable for Business Intelligence, analytics, and visualization purposes (Muhammed, 2024). The dataset includes historical data on pizza sales, which can be used to represent the performance of a pizza restaurant as a business. In this report, the pizza sales dataset will be used to evaluate the performance of The Pizza Factory by creating an interactive dashboard using **Microsoft Power BI**.

3.2 Dataset Characteristics

The dataset is stored by means of **Comma-Separated Values (CSV)** and consists of **48,620 sales records**. Each record contains **12 fields (Attributes)** describing different aspects of the sold pizza. The attributes found in the set represent all of the features of the restaurant's sales process from the information about the customer to the specific size and price of pizza. The attributes also show how many items were ordered in each transaction and when the orders were placed. This allows to study the process of sales in-depth and analyze customer behavior as well as the popularity of certain pizza types and sizes.

3.3 Data Types

This dataset consists of several types of data that support different forms of analysis:

Table 3.1 Data Types

Identifier Data:	<u>Pizza ID</u>, <u>Order ID</u>, Pizza Name ID
Numerical Data:	Quantity, Unit Price, Total Price
Categorical Data:	Pizza Category, Pizza Size, Pizza Name, Pizza Ingredients
Date and Time Data:	Order Date, Order Time

From the combination of these data types, it enables us to understand Business Intelligence Analysis by allowing numerical calculations, trend analysis, customer behavior analysis, and product performance evaluation

3.4 Purpose of The Dataset

The main purpose of this dataset is to create sales information that may be helpful to management in order to make the best business decisions by using **Business Intelligence**. For example, by watching over the sales, the management will see how the whole business operates, what kind of pizzas is the most and the least popular, how much money the business makes, when to expect the highest pizza sales, what customers like the most. All of these factors will help **The Pizza Factory** to operate more productive and efficient business and satisfy all of its stakeholders.

3.5 Data Quality

Before carrying out any analysis, it is important to allow the data set to go through the process of data cleaning in **Power Query of Microsoft Power BI**. In this case, the data cleansing process included the removal of missing data, checking for consistency, cleaning of data types, and elimination of additional errors from the data set to ensure that the data is amenable to analysis and facilitates quality results and decisions. The data cleaning process is important because it ensures the accuracy and quality of the data under analysis.

3.6 Expected Insights from Dataset

The cleaned data is expected to produce meaningful business insights from Power BI dashboards. The analysis will focus on the total sales revenue, total orders, average order value, the most popular pizzas, sales by pizza categories and sizes, the busiest ordering hours, and monthly sales trends. The findings will be used to inform operational managers and senior management to improve operations, boost profits, and make better business decisions in The Pizza Factory.

4.0 DATA CLEANING PROCESS

The dataset of pizza sales was first cleaned and prepared in Power Query Editor of Microsoft Power BI before data analysis and Business Intelligence (BI) dashboards were developed. Data Cleaning is an important part of the Business Intelligence process as it makes sure that the information set is accurate, consistent, and appropriate for analysis. In this phase, multiple data preparation methods were used to enhance the quality of the data set before it's added to Power BI for visualization.

4.1 Data Ingestion and Environment Setup

To securely load the raw dataset into the analytics environment without altering the source file. The **pizza_sales_dataset.xlsx** file, containing 1,000 randomized records from May 2026, was imported into Power BI Desktop. The data was immediately routed into the Power Query Editor to begin the transformation pipeline. Using Power Query, we ensure that all data cleaning steps are recorded as a repeatable script, preserving the integrity of the raw data while allowing for automated updates in the future.

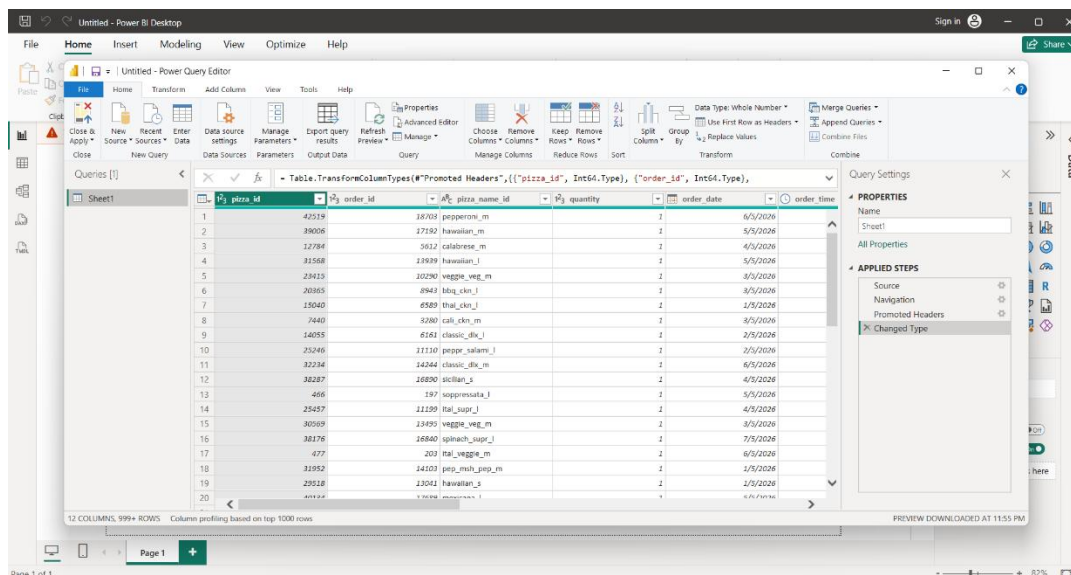


Figure 4.0.1 Raw dataset was successfully loaded into the Power Query Editor environment

4.2 Data Inspection and Null Value Handling

This process is to verify data completeness and eliminate incomplete records that could skew financial or operational metrics. We have conducted a column quality inspection across the dataset. Utilized the column filter tool on primary fields such as unit price and quantity) to scan for missing values. The inspection confirmed the dataset was 100% complete. **No "(null)" or "(blank)" entries** were found, meaning all 1,000 records were retained for analysis.

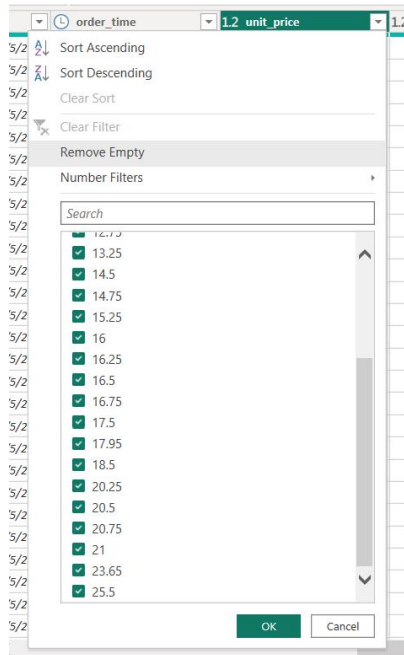


Figure 4.0.2 Verifying the absence of any (null) or (blank) values.

4.3 Standardization of Data Types

We strictly define the schema of the dataset so Power BI can accurately process chronological and financial aggregations. We have reclassified the data types of key operational columns. The order date column was converted from standard text to a Date type. Financial metrics, specifically unit price, were converted to a Fixed Decimal Number type currency format.

The time intelligence functions, like week-over-week comparisons, cannot operate unless dates are explicitly categorized as Date objects. Similarly, fixed decimal types prevent rounding errors during large financial summations.



order_date
6/5/2026
5/5/2026
4/5/2026
5/5/2026
3/5/2026

Figure 4.0.3 Data type actively selected from the transformation



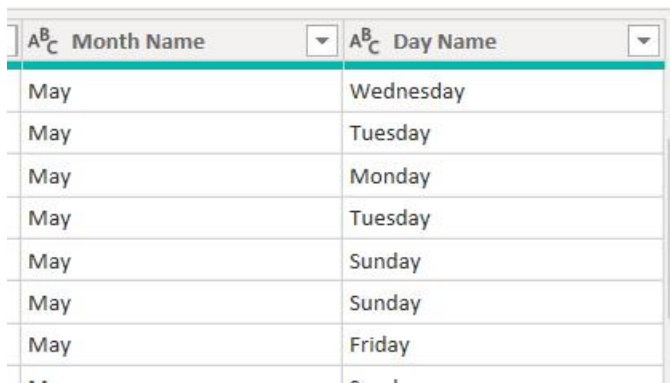
\$ unit_price
12.50
13.25
16.25
16.50
16.00

Figure 4.0.4 Data type actively selected from the transformation

4.4 Feature Engineering: Time Series (Month & Day)

This process is to generate categorical variables that allow the dashboard to filter and visualize trends based on specific days of the week or months. We managed to engineer two new columns directly from the order date field using Power Query's built-in Date extraction tools

Extracted the "Name of Month" (yielding "May") and the "Name of Day" e.g., "Monday", "Friday" into standalone columns at the end of the dataset.



The image shows a screenshot of a data table with two columns: 'Month Name' and 'Day Name'. Both columns have a dropdown arrow icon in the header. The table contains several rows of data, all with 'May' in the 'Month Name' column and various days of the week in the 'Day Name' column. The rows are: May Wednesday, May Tuesday, May Monday, May Tuesday, May Sunday, May Sunday, and May Friday. The table is part of a larger application, as indicated by the 'ABC' icons in the column headers.

ABC Month Name	ABC Day Name
May	Wednesday
May	Tuesday
May	Monday
May	Tuesday
May	Sunday
May	Sunday
May	Friday

Figure 4.0.5 "Month Name" and "Day Name" columns successfully appended to the dataset.

4.5 Feature Engineering: Financials (Revenue)

This is the crucial step in our research to establish the primary Key Performance Indicator (KPI) for the company's financial dashboard by calculating the total monetary value of every individual order. So, we created a Custom Column named Revenue.

We applied the mathematical formula **[quantity] * [unit price]** to calculate the total transaction value. The resulting column was immediately standardized to a Fixed Decimal Number data type to align with financial reporting standards.

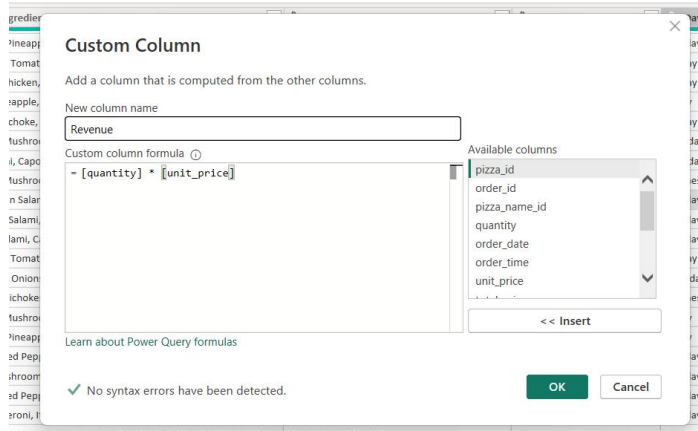


Figure 4.0.6 Displaying mathematical formula used to engineer the Revenue metric.

4.6 Finalization and Deployment

Lastly , to lock in the transformation pipeline and deploy the cleaned data model to the visual dashboard interface we executed the "Close & Apply" command in Power Query. So , power BI successfully processed the applied steps (ingestion, schema definition, and feature engineering) and loaded the optimized, 1,000-row dataset into the primary workspace, ready for visual modeling.

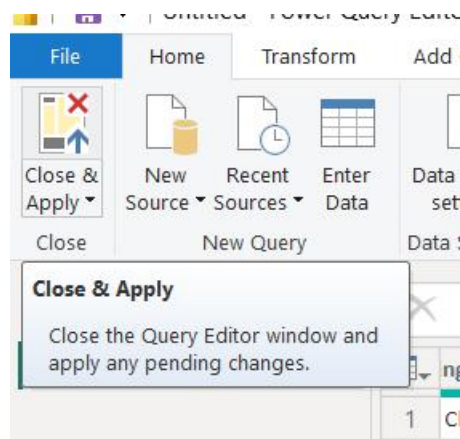


Figure 4.0.7 Close & Apply command to process the applied steps.

5.0 DATA ANALYSIS AND DASHBOARD DEVELOPMENT

5.1 Dashboard Visualization (Pizza Sales Dashboard)

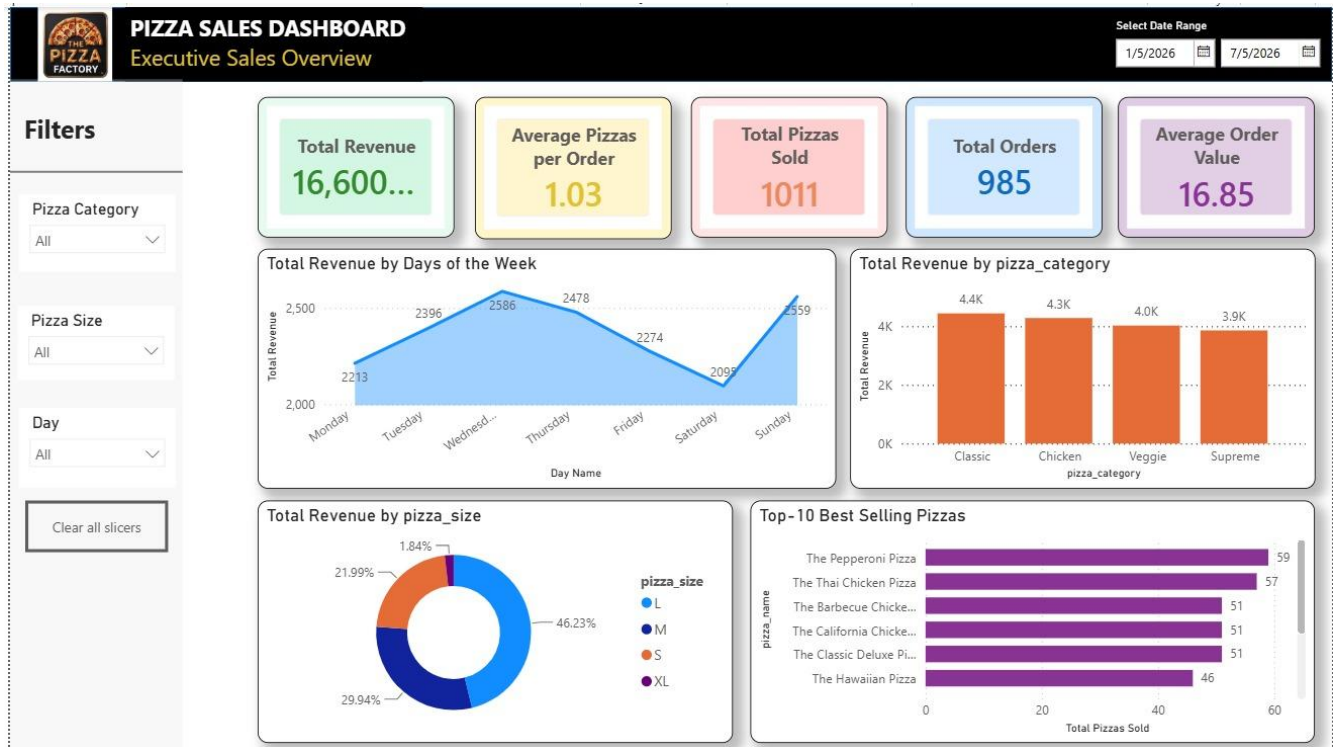


Figure 5.0.1 Pizza Interactive Dashboard

Microsoft Power BI has been used to create the Pizza Sales Dashboard to show an overview of The Pizza Factory's sales. It integrates essential business indicators and interactive graphics into a unified dashboard, enabling management to streamline their monitoring process and make well-informed decisions.

There are five **KPI cards**, a **line chart**, a **clustered column chart**, a **donut chart** and a **horizontal bar chart** on the dashboard to analyze the revenue, sales trends, product categories, pizza sizes, and best-selling pizzas (Microsoft, 2026) . Interactive slicers for Pizza Category, Pizza Size, Day and Date Range allow users to filter the data and discover particular business insights

Overall, the dashboard is a Business Intelligence reporting tool which displays sales information in an easy-to-follow, interactive way, supporting the Operational managers to make data-driven decisions to improve business performance.

5.1.1 Key Performance Indicator (KPI Cards)



Figure 5.0.2 KPI Cards

The most important business metrics are displayed at the top of the dashboard in the Key Performance Indicator (KPI) cards which give a quick overview of the overall sales performance of The Pizza Factory. These indicators help management to determine the performance of the business at a glance without having to go through a lot of sales details.

The Total Revenue KPI reflects the overall company's revenue in the chosen time frame, helping management track the company's financial performance. The **Total Orders KPI** shows the total volume of customer orders filled, and the **Total Pizzas Sold KPI** shows the total number of pizzas sold, giving insights into customer demand and sales volume. The **Average Order Value** is the average amount spent per order from the customer, which helps management analyses the customer spending pattern. Finally, **Average Pizzas per Order** reflects the average number of pizzas bought per order, which could be utilized to assess purchasing patterns and explore the opportunity of product bundling or promotional offers.

These KPI cards work well together to give an overview of the most important business performance indicators for the company and help operational managers to track the performance of their sales, analyses the customer purchase behavior and make informed decisions that help them grow and operate their business more effectively.

5.1.2 Revenue by Days of the Week (Line Chart)

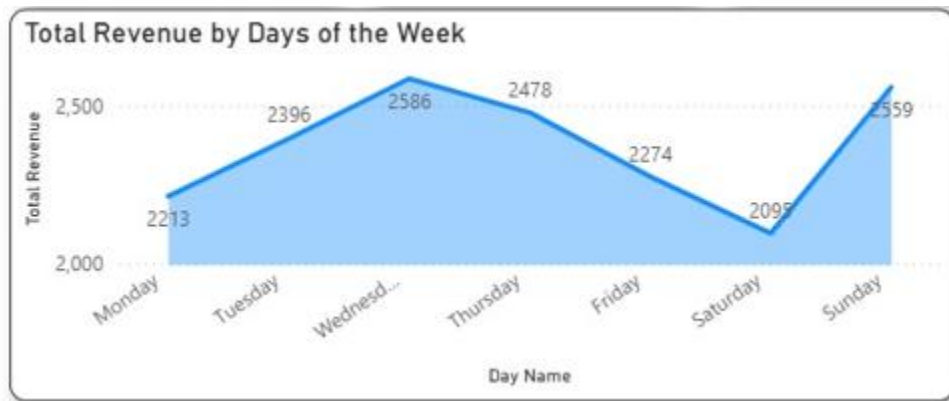


Figure 5.0.3 Line Chart

The Total Revenue by Days of the Week line chart provides a breakdown of sales across the week's days, which helps management observe the sales patterns and customer buying behavior each day of the week. Wednesday collected the most money from the dashboard with **2,586**, while Sunday nearly tied with it at **2,559** and Thursday was next at **2,478**. Conversely, on Saturday the least sales activity occurred with lower sales (2095) in correlation with the revenue collected.

This visualization assists operational managers in recognizing when it is a peak and an off-peak business day, to help them run their businesses with optimum staffing, inventory planning and promotion plans. For instance, extra staff can be assigned during days of particular high turnover, like Wednesday and Sunday, while special promotions or offers could be facilitated during days of decreased turn, for instance Saturday to energize buyer activity and gather customers.

5.1.3 Revenue by Pizza Category (Column Chart)

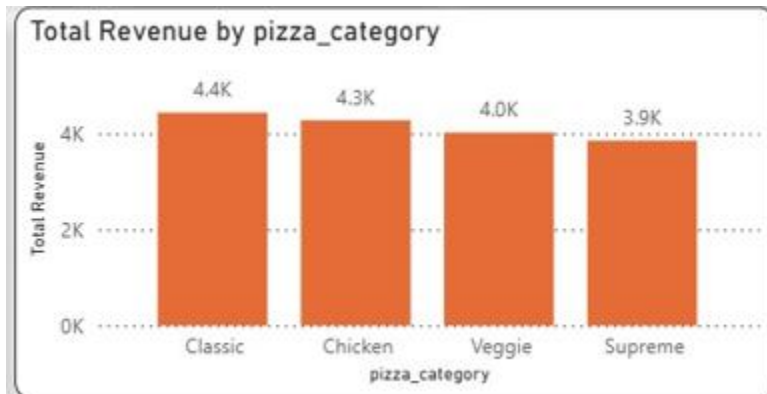


Figure 5.0.4 Column Chart

The Total Revenue by Pizza Category column chart lets management look at the revenue generated for each pizza category, enabling them to compare the financial results of the various product groups. The revenue analysis shows the Classic, Chicken (**4.3K**), Veggie (**4.0K**) and Supreme (**3.9K**) categories as the top ones, based on the dashboard data. While the Classic segment's revenue is fairly modest in comparison, it is still the top performer.

This visualization can help operational managers decide which are the best pizza types and plan the menus, manage the inventory and promotional actions accordingly. The analysis can help determine which categories are more successful, like Classic, and how they can be targeted during marketing campaigns, and which are less successful, like Supreme, that may need specific promotions or menu tweaks to boost customers' appetite and overall success.

5.1.4 Revenue by Pizza Size (Donut Chart)

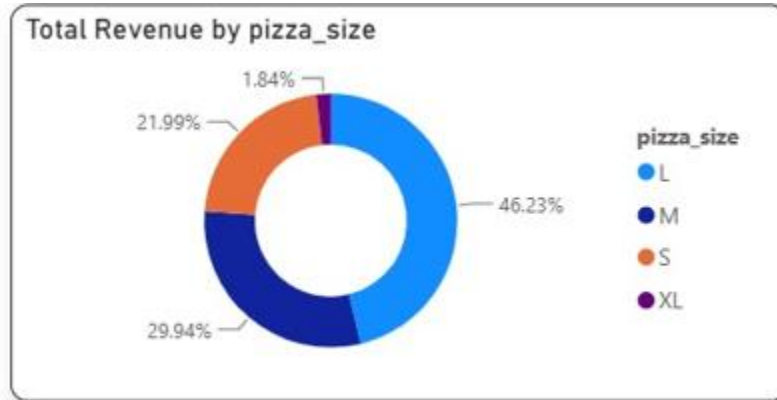


Figure 5.0.5 Donut Chart

The Donut chart “Total Revenue by Pizza Size” represents the percent contributed from each pizza size to the company's total revenue. **The Large (L)** pizzas had the largest revenue percentage which was 46.23%, **Medium (M)** pizzas 29.94% and **Small (S)** pizzas 21.99%, as shown on the dashboard. The sales of the smallest sized **pizzas (Extra Large XL)** were at 1.84%.

The merits of this visualization are to identify the preferred pizza sizes, which tells you the preferences of your customers. From the data it is inferred that whereas Large pizzas are highly demanded, XL pizzas are sold relative less as the customers are less demanding about getting large size of a pizza. With these data insights, operational managers can more effectively optimize inventory, as well as optimize pricing and develop specific promotions more focused on customer preference to maximum value.

5.1.5 Top 10 Best-Selling Pizzas (Bar Chart)

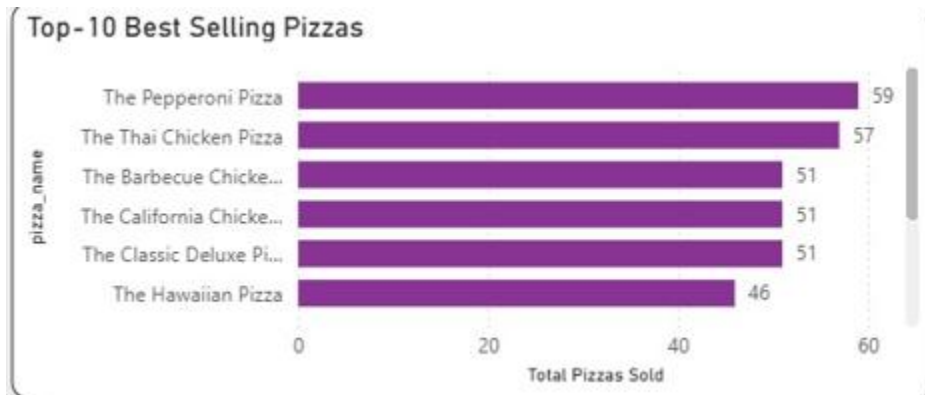
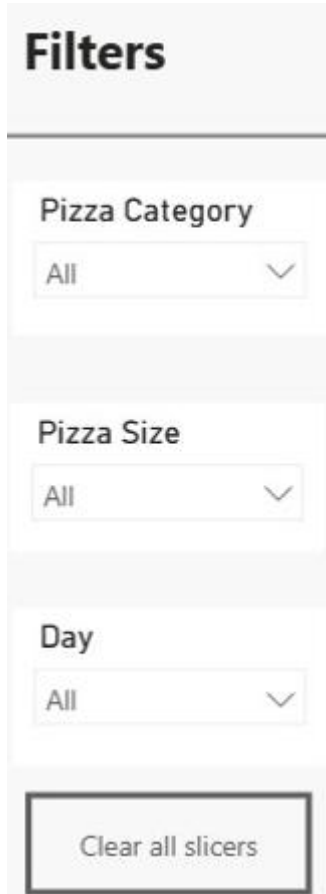


Figure 5.0.6 Bar Chart

The horizontal bar chart 'Top 10 Best Selling Pizzas' sorts the totals of the number of pizza products sold so that management can easily capture the bestselling menu items. The number of pizzas sold on the dashboard is used for the analysis. The graph shows the sales of **The Pepperoni Pizza** followed by **The Thai Chicken Pizza** based on the number of pizzas sold on the dashboard. The **Barbecue Chicken Pizza**, **The California Chicken Pizza** and **The Classic Deluxe Pizza** all had 51 sales each, and **The Hawaiian Pizza** had 46 sales.

This type of visualization can help to identify customers' preferences and product performance. Taking note of the most popular pizza items through the identification process can help operational managers to ensure that they have adequate supplies of popular ingredients, optimize their menus, and engage in marketing campaigns to promote the most well-known products. This data is also used to help with production planning and inventory management, which helps The Pizza Factory run efficiently and maximize their sales revenue.

5.1.6 Interactive Filters (Slicers)



Filters

Pizza Category
All ▼

Pizza Size
All ▼

Day
All ▼

Clear all slicers

Figure 5.0.7 Slicers

The Dashboard has interactive slicers for Pizza Category, Pizza Size, and Day, and a clear button to make cleaning up all the slicers easy for the user to use and explore the data. Such filters allow users to adjust the dashboard based on their specific criteria and the visualizations and KPI cards automatically update when they change.

By using the Pizza Category slicer, management can make comparisons on the sales performance of each pizza category. By using Pizza Size slicer management can make comparisons of the revenue generated from each size of pizza. Sales can be viewed in the Day slicer with a specific day of the week. There is also the Clear All Slicers button option to easily undo the multiple filters applied and get the dashboard back to its default state.

5.1.7 Date Range Filter (Date Range Slicer)

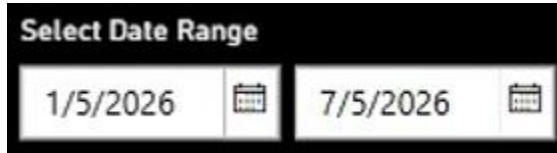


Figure 5.0.8 Date Range Filtler

The Date Range filter allows to analyses sales performance in a period of time as per the user's selection by using the Date Start and Date End filters. If the date range is defined, then all the KPI cards and visualization the dashboard will also show only the data in that range.

It is a great advantage for operational managers to view performance of sales over time, compare to previous periods of time to determine if a short-term trend exists, and then ascertain the impact of successful promotions, seasonal events etc. on performance. The Date Range filter allows you to filter and get detailed analysis through different time periods that will help you to monitor business performance more accurately and help management to make quick decisions based on the analysis.

6.0 DATA MINING TECHNIQUE

6.1 Selected Data Mining Technique

The objective of this project is to employ the data mining technique Time Series Analysis with the objective of analyzing the historical sales data of The Pizza Factory (Lee & Kim, 2015). Time Series Analysis is a data mining technique for analyzing data that is gathered over time to either uncover patterns, trends and fluctuations in business activity or to prepare it for analysis by other techniques. In Business Intelligence (BI), this carries an advantage for organizations which can use it to track changes in sales, predict future sales and help guide data-driven decisions from past information.

The sales information includes **Order Date, Day Name, Revenue, Quantity Sold, Pizza Category and Pizza Size** attributes this enables sales performance to be analyze over the course of time. The time series analysis on the line chart of Total Revenue by Days of the Week highlights the daily occurrences of revenue, and enabled the identification of days with the highest or lowest revenue generation as well. These trends can be analyses and understood by management for customer purchasing behavior and to examine operational performance.

The model chosen was **Time Series Analysis**, as it is applicable to the analysis of sales transaction which takes place at constant time intervals. Whereas **Classification** or **Clustering Methods** emphasize the grouping and prediction of classes, Time Series Analysis shows the importance of recognizing temporal patterns and trends which can aid businesses in their sales forecasting and operational planning.

6.2 Justification for Choosing Time Series Analysis

The decision to use **Time Series Analysis** was made because it was deemed to be a very effective method to achieve the goals of this **Business Intelligence project**, as this technique can convert past sales data into more business-intelligent information. By applying the technique, management can better assess revenues from day to day and recognize whether there are any common sales cycles that could affect business functions.

The dashboard shows that sales revenue fluctuates from day to day for example during the week some days bring in more money than the others. Such information enables the operational managers to optimize their staff scheduling, inventory management, and promotional planning based on customer needs. For instance, more personnel and items may be made use of during busy selling seasons and promotional offers can be put into place during less busy periods to drive consumers to the store and make extra sales.

Moreover, Time Series Analysis enables strategic decision making in terms of historical trends, which can be the basis of forecasting sales in the future. By studying historical sales data, The Pizza Factory can predict customer orders, prevent stockouts, minimize food waste and enhance operational efficiency.

6.3 Justification of BI Reporting Solution and Dashboard Design

The use of an interactive dashboard in the Microsoft Power BI, which enables management to track this business performance clearly and interactively, has been developed as the main Business Intelligence reporting solution. This dashboard presents several performance parameters in one screen, so managers can efficiently study sales data without scanning through a great deal of raw data from transactions. The KPI cards were chosen to give a quick overview of relevant business metrics: Total Orders, Total Order Value, Total Pizzas Sold, Average Order Value, Average Pizzas per Order. These measures enable management to make a "snapshot" assessment of overall business performance.

A line chart was selected to illustrate Time Series Analysis to show revenue over the various days of the week. This is a visual which will be useful to pinpoint the time periods where sales have increased and decreased and contribute to planning business activities. The clustered column chart shows the revenue from individual pizza types, allowing management to assess the financial success of the various product lines, and determine which product lines make the largest contribution to the company's income. The donut chart helps depict the percentage of sales by pizza size, and gives customers some insights on customer purchasing. Knowing which sizes are biggest revenue generators will help management plan about their pizza inventory and pricing plans.

The bar graph below shows some pizzas being ranked in order of their quantity sold. The visual aids in determining the most popular items, allowing the company to refine their menu, make sure they have plenty of ingredients, and create marketing strategies specifically for favored menu items. Lastly, there are interactive Slicers on the dashboard: Pizza Category, Pizza Size, Day, and Date Range. These filtering tools enable managers to analyze different business situations by taking the various changes at each visualization dynamically in order to consider different criteria. The interactive elements make the dashboard more user-friendly and make it easier to explore sales data in depth. Overall, it can be concluded that the dashboard designed is effective in presenting key business information; it is simple, interactive and visually organized, with the capacity to enable operational managers to take timely decisions to enhance business performance, based on the facts presented.

7.0 ANALYSIS AND FINDINGS

The dashboard analysis offers good information about the performance of sales of The Pizza Factory, offering important business parameters and user's purchasing trends. The analysis of revenue, customer orders, product performance and sales trends can help management identify areas of strength, opportunities for improvement, and opportunities to improve operational efficiency.

7.1 Overall Sales Performance

The dashboard shows a total revenue for The Pizza Factory of about **16.6K** in the period selected, averaging to **1.1K** per order and **1.6K** per pizza sold, as well as **985** orders from customers and 1,011 pizzas were drawn from the inventory. An average transaction value is **16.85** and the average number of pizzas per order is **1.03**. These numbers suggest that customers are likely to place single item orders rather than buying more than one pizza in a sitting, as the average number of items bought per transaction there.

The **KPIs** compiled in the results show a stable business performance, and enable management to have a clear idea about the business sales activities. By regularly checking these indicators, managers can assess the overall performance and track changes in customer buying behavior over time.

7.2 Sales Performance by Day of the Week

When analyzing revenue by day of the week, there are clear differences observed in buying patterns among customers throughout the **week**. Wednesday had the **most in revenue** followed by Sunday (**2,559**) then Thursday (**2,478**). Saturday produced the least of the revenues (**2,095**).

The results indicated that the high activity level occurred more in the middle of the week and on Sunday and low sales occurred mainly on Saturdays. By analyzing these sales trends, management can make informed decisions on workforce allocation, restocking during slow sales times and using sales promotions on less busy days to leverage extra sales.

7.3 Revenue by Pizza Category

The **Classic pizza** category had the **highest revenue** of about **4.4K**, followed by **Chicken** at **4.3K**, **Veggie** at **4.0K** and **Supreme** at **3.9K**. While there may not be a huge disparity in revenue between one category and another, Classic continues to bring in the most revenue for the company.

This suggests that the market demand is fairly evenly spread among the available categories of pizza, but there is some more demand for Classic pizzas. Based on this information, management can prioritize the stocking of popular categories and to strategize the promotion and sales efforts on lower performing categories to boost overall sales.

7.4 Customer Preference by Pizza Size

The donut chart shows that Large pizza size is the favorite among customers as it accounts for **46.23%** of total revenue. This pizza size's sales are the biggest contributing factor compared to other sizes. Medium pizza size comes second with **29.94%** of total revenue, followed by Small pizza size with **21.99%**. Finally, there is only **1.84%** of total revenue from Extra Large pizza size sales. The chart's data suggests that customers prefer to buy bigger pizza sizes as they seem to be more cost-efficient compared to smaller ones. However, Extra Large pizza size is rarely purchased, possibly, due to its lower demand or higher price. Thus, the management can optimize pizza size offerings based on customer demand and assess prices in accordance with pizza sizes.

7.5 Best Selling Pizza Products

As per the provided dashboard, it is determined that The Pepperoni Pizza is the most selling product, with **59 sold** pieces, and the second pizza in terms of popularity is The Thai Chicken Pizza, with **57 sold** pieces. The other popular pizzas include The Barbecue Chicken Pizza, The California Chicken Pizza, and The Classic Deluxe Pizza, with **51 sold** pieces each, while The Hawaiian Pizza is at the sixth position with **46 sold** pieces.

Hence, as it can be seen from the data, there are several popular pizzas for which the company should ensure the steady supply to meet the demand. Moreover, the company could implement some marketing strategies such as special offers for the most popular pizzas to increase the revenues and promote their sales.

8.0 RECOMMENDATIONS

Based on the Business Intelligence analysis results presented in the **Power BI dashboard**, this memo will provide recommendations for further improvements of The Pizza Factory's operations based on the data and information retrieved. Firstly, The Pizza Factory should consider optimizing the inventory based on the sales information retrieved from the dashboard. According to the dashboard, **The Pepperoni Pizza** and **The Thai Chicken Pizza** are the **most popular** menu items retrieved since both have consistently **high sales** throughout the analyzed period. Therefore, the management should ensure that there is an adequate stock of ingredients for these menu items since there is a high demand for them.

The management should consider aggressive marketing and promotional campaigns for the menu items with the lowest sales and revenues. It is essential since the dashboard suggests that all categories of pizzas have almost the same impact on the overall revenue, except Supreme, which appears to be the least selling category. Promoting the low-selling menu items and providing special offers or discounts for buying a combination of pizzas from different categories may positively affect the company's sales and revenues.

The Pizza Factory's management should optimize the staff schedule based on the sales information retrieved from the dashboard. Notably, the dashboard shows that on average, the highest sales revenues are registered on Wednesdays and Sundays, while the lowest sales revenue is registered on Saturdays. Additionally, staff scheduling helps maintain the quality of customer service since the number of employees corresponds to the level of demand for the company's services.

Finally, The Pizza Factory should employ Business Intelligence on a regular basis to ensure the company's steady growth and sustainable development in the highly competitive food and beverage industry. It is recommended that the management uses the Power BI dashboard more frequently to retrieve vital information and data on the company's performance, which will aid in the efficient decision-making process and help the management monitor the performance of the employees, the sales of the company, and its operations in general.

9.0 CONCLUSION

This project was a great opportunity to gain experience in one of the essential aspects of modern business, namely Business Intelligence. A dashboard was built in Microsoft Power BI to help The Pizza Factory analyze the company's sales. First, a pizza sales dataset was imported into the software and prepared for analysis using the Power Query tool. It was found that the data needed some cleaning, so several queries were created to complete this task. After the data had been processed, a variety of charts and graphs were created to present this information in a compelling way. The final step was to apply all these visualizations in Microsoft Power BI to create an informative and attractive dashboard that would allow The Pizza Factory management to keep track of the company's sales, observe customers' purchasing behavior see what pizza sizes and products are the most popular, detect sales trends, and much more.

The analysis of the pizza sales dataset allowed us to identify a few essential insights that could help The Pizza Factory management improve the overall performance of the company. In particular, the company's sales, the most popular pizza category, the size customers prefer, the most selling pizza products, and the revenues of the pizza shop on different days of the week were analyzed. It is evident that Business Intelligence can be an extremely effective tool for any modern company, as it allows businesses to track their progress, assess their performance, discover business trends, and predict future demands.

Moreover, the Time Series Analysis data mining technique was used in this project, which demonstrates how analytical techniques can help businesses make efficient decisions and become more competitive in the modern business world. Finally, a set of evidence-based and data-driven recommendations was developed for The Pizza Factory management, which should allow them to improve the company's performance.

To conclude, this project shows how crucial Business Intelligence is for any modern company, as it enables businesses to turn data into information, which in turn can help organizations improve their performance. The dashboard that was created in Microsoft Power BI allows The Pizza Factory management to have an insight into the company's sales and, thus, makes them more competitive.

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